4. Testing

4.1 Testing plan and results **Normal data**

Test no.	Description	Input data	Expected result	Actual result	P / F
1	One small pizza, one topping, collection	Pizzas: 1 Pizza 1: size 1, 1 topping Delivery: false	Pizza 1 £4.00. No discount. Order Total = £4.00	Pizza cost shown while ordering: £4. Discount not earned. Order Total = £4.00	Pass
2	One medium pizza, two toppings, with delivery	Pizzas: 1 Pizza 1: size 2, 2 toppings Delivery: true	Pizza 1 £6.85. No discount; delivery £2.50 added. Order Total = £9.35	Pizza cost shown while ordering: £6.85. Discount not earned. Order Total = £9.35	Pass*
3	Two different pizzas, no discount	Pizzas: 2 Pizza 1: size 3, 3 toppings Pizza 2: size 2, 1 topping Delivery: false	Pizza 1 £9.15; Pizza 2 £6.25. No discount. Order Total = £15.40	Pizza costs shown while ordering: £9.15, £6.25. Discount not earned. Order Total = £15.40	Pass
4	Discount earned (subtotal over £20)	Pizzas: 4 Every pizza: size 2, 1 topping Delivery: false	Each pizza £6.25. Discount applied (£2.50). Order Total = £22.50	Pizza costs shown while ordering: £6.25 each. Discount amount = £2.50. Order Total = £22.50	Pass
5	Discount and delivery together	Pizzas: 3 Every pizza: size 3, 2 toppings Delivery: true	Each pizza £8.50. Discount applied (£2.55); delivery £2.50 added. Order Total = £25.45	Pizza costs shown while ordering: £8.5 each. Discount amount = £2.55. Order Total = £25.45	Pass*
6	Bill shows every required item (use the order from test 5)	Same as test 5	Customer name, address and phone. Each pizza: standard £7.15 + toppings £1.35 = £8.50. Subtotal before discount and delivery £25.50. Discount £2.55. Delivery charge £2.50. Final total £25.45.	Customer details shown (name, address, phone). Pizza costs shown while ordering only (£8.5 each), not on the bill. Bill shows: discount amount £2.55 and Order Total £25.45. Missing from bill: standard and topping	Fail

				cost, subtotal before discount, delivery charge.	

Boundary data

Test no.	Description	Input data	Expected result	Actual result	P / F
7	Minimum pizzas (1)	Pizzas: 1 Pizza 1: size 3, 1 topping Delivery: false	Pizza 1 £7.90. No discount. Order Total = £7.90	Pizza cost shown while ordering: £7.9. Discount not earned. Order Total = £7.90	Pass
8	Maximum pizzas (6)	Pizzas: 6 Every pizza: size 1, 1 topping Delivery: false	Each pizza £4.00. Discount applied (£2.40). Order Total = £21.60	Pizza costs shown while ordering: £4 each. Discount amount = £2.40. Order Total = £21.60	Pass*
9	Pizza count just below minimum	Pizzas: 0, then 1 Pizza 1: size 1, 1 topping Delivery: false	Error, must order minimum 1 and maximum 6 pizzas, then asks again. 1 pizza accepted. Order Total = £4.00	Error shown once, then asked again. 1 pizza accepted. Discount not earned. Order Total = £4.00	Pass
10	Pizza count just above maximum	Pizzas: 7, then 1 Pizza 1: size 1, 1 topping Delivery: false	Error, must order minimum 1 and maximum 6 pizzas, then asks again. 1 pizza accepted. Order Total = £4.00	Error shown once, then asked again. 1 pizza accepted. Discount not earned. Order Total = £4.00	Pass
11	No extra toppings (0)	Pizzas: 1 Pizza 1: size 1, 0 toppings Delivery: false	Topping cost £0.00. Pizza 1 £3.25. No discount. Order Total = £3.25	0 toppings charged £0.75 (pizza cost shown as £4, expected £3.25). Discount not earned. Order Total = £4.00	Fail
12	Four extra toppings (start of '4 or more')	Pizzas: 1 Pizza 1: size 2, 4 toppings Delivery: false	Pizza 1 £8.00. No discount. Order Total = £8.00	Pizza cost shown while ordering: £8. Discount not earned.	Pass

				Order Total = £8.00	
13	Subtotal exactly £20.00 (no discount)	Pizzas: 5 Every pizza: size 1, 1 topping Delivery: false	Each pizza £4.00. No discount. Order Total = £20.00	Pizza costs shown while ordering: £4 each. Discount not earned. Order Total = £20.00	Pass
14	Subtotal just over £20 (£20.10)	Pizzas: 3 Pizza 1: size 1, 2 toppings Pizza 2: size 2, 3 toppings Pizza 3: size 2, 4 toppings Delivery: false	Pizza 1 £4.60; Pizza 2 £7.50; Pizza 3 £8.00. Discount applied (£2.01). Order Total = £18.09	Pizza costs shown while ordering: £4.6, £7.5, £8. Discount amount = £2.01. Order Total = £18.09	Pass*
15	Delivery must not count towards the £20 limit	Pizzas: 3 Every pizza: size 2, 1 topping Delivery: true	Each pizza £6.25. No discount; delivery £2.50 added. Order Total = £21.25	Pizza costs shown while ordering: £6.25 each. Discount not earned. Order Total = £21.25	Pass
16	Pizza size just outside range	Pizzas: 1 Pizza 1: size 0, then 4, then 1; 1 topping Delivery: false	Error: size must be within range (each time), then asks again. Size 1 accepted. Order Total = £4.00	Error shown twice, then asked again. Size 1 accepted. Discount not earned. Order Total = £4.00	Pass
17	Negative number of toppings	Pizzas: 1 Pizza 1: size 1; toppings -1, then 1 Delivery: false	Error, you can't have negative toppings, then asks again. 1 topping accepted. Order Total = £4.00	Error shown, but 'Ok you've selected -1 extra toppings' is also printed. Asked again, 1 topping accepted. Order Total = £4.00	Pass

Extreme data

Test no.	Description	Input data	Expected result	Actual result	P / F
18	Pizza count far outside range	Pizzas: 100, then -5, then 1 Pizza 1: size 1, 1 topping	Error message each time, then asks again. 1 pizza accepted.	Error shown twice, then asked again. 1 pizza accepted.	Pass

		Delivery: false	Order Total = £4.00	Discount not earned. Order Total = £4.00	
19	Very large number of toppings	Pizzas: 1 Pizza 1: size 1, 50 toppings Delivery: false	Topping cost capped at £2.50. Pizza 1 £5.75. No discount. Order Total = £5.75	Pizza cost shown while ordering: £5.75. Discount not earned. Order Total = £5.75	Pass
20	Pizza size far outside range	Pizzas: 1 Pizza 1: size 99, then 1; 1 topping Delivery: false	Error: size must be within range, then asks again. Size 1 accepted. Order Total = £4.00	Error shown once, then asked again. Size 1 accepted. Discount not earned. Order Total = £4.00	Pass
21	Blank customer details	Name: press Enter with nothing typed, then Sam Taylor Address: spaces only, then 1 High Street Phone: 07123456789 Pizzas: 1; size 1, 1 topping; Delivery: false	'This cannot be blank, try again!' shown for the blank name and the spaces-only address. Asks again until real details are entered. Order continues, Order Total = £4.00	'This cannot be blank, try again!' shown twice (blank name, spaces-only address). Details accepted once valid. Order continues. Order Total = £4.00	Pass
22	Largest possible order	Pizzas: 6 Every pizza: size 3, 10 toppings Delivery: true	Each pizza £9.65. Discount applied (£5.79); delivery £2.50 added. Order Total = £54.61	Pizza costs shown while ordering: £9.65 each. Discount amount = £5.79. Order Total = £54.61	Pass*
23	Text or decimal for number of pizzas	Pizzas: abc, then 2.5, then 1 Pizza 1: size 1, 1 topping Delivery: false	Inputs rejected and asked again (no crash, no calculation). 1 pizza accepted. Order Total = £4.00	Not traced: depends on how Flowgorithm handles text or a decimal typed into an Integer input. Run to record.	To run
24	Invalid delivery answer	Pizzas: 1; size 1, 1 topping Delivery: maybe, then false	'maybe' rejected and asked again (no crash). false accepted. Order Total = £4.00	Not traced: depends on how Flowgorithm handles an invalid Boolean input. Run to record.	To run